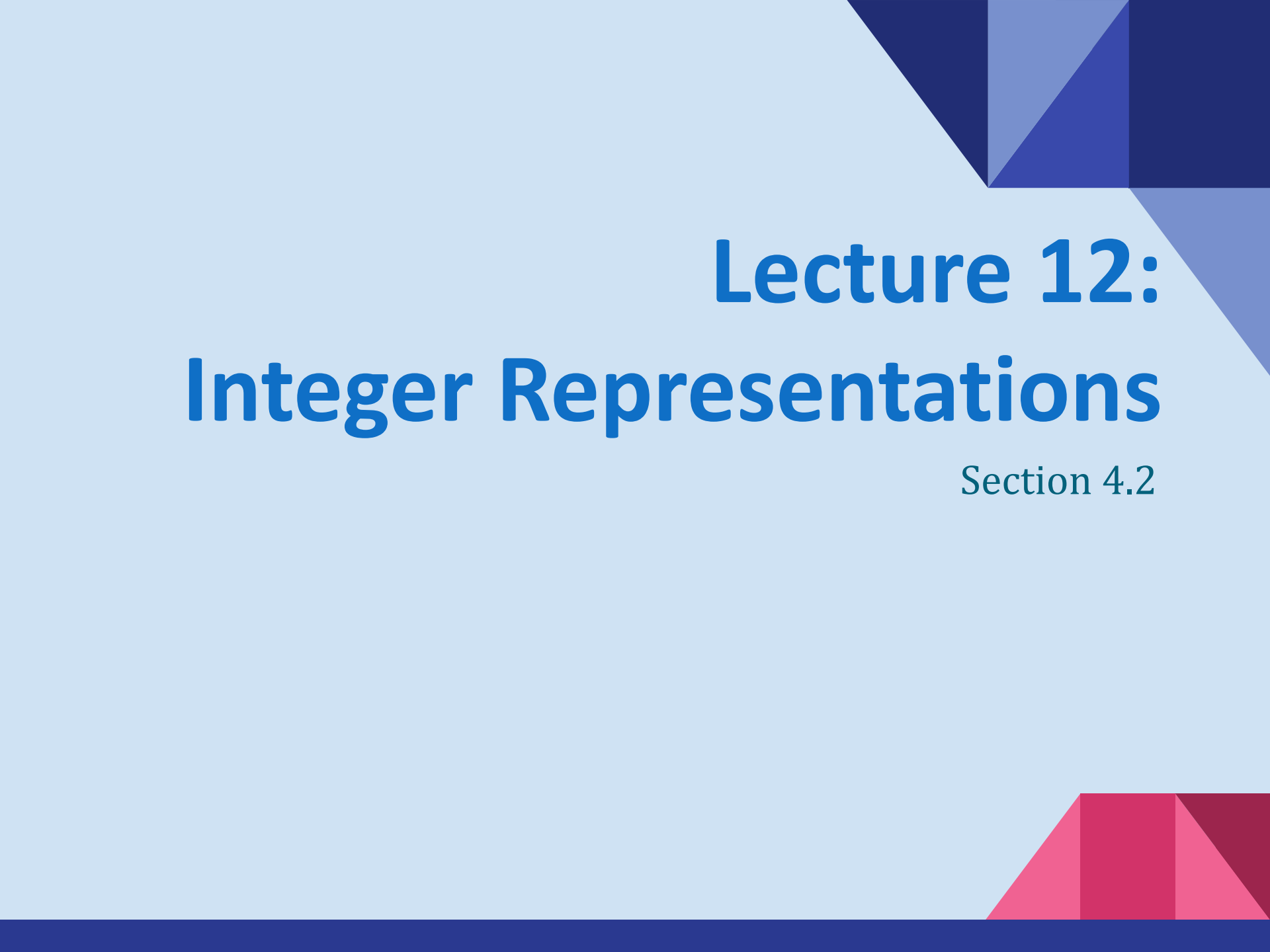


The slide features a light blue background. In the top right corner, there is a decorative arrangement of overlapping triangles in various shades of blue and purple. In the bottom right corner, there is a similar arrangement of overlapping triangles in shades of pink and red. The main title is centered on the slide.

# Lecture 12: Integer Representations

Section 4.2

# Section Summary

- Integer Representations
  - Base  $b$  Expansions
  - Binary Expansions
  - Octal Expansions
  - Hexadecimal Expansions
- Base Conversion Algorithm
- Algorithms for Integer Operations

# Representations of Integers

- In the modern world, we use *decimal*, or *base 10, notation* to represent integers. For example when we write 965, we mean  $9 \cdot 10^2 + 6 \cdot 10^1 + 5 \cdot 10^0$ .
- We can represent numbers using any base  $b$ , where  $b$  is a positive integer greater than 1.
- The bases  $b = 2$  (*binary*),  $b = 8$  (*octal*), and  $b = 16$  (*hexadecimal*) are important for computing and communications
- The ancient Mayans used base 20 and the ancient Babylonians used base 60.

# Base $b$ Representations

- We can use positive integer  $b$  greater than 1 as a base, because of this theorem:

**Theorem 1** : Let  $b$  be a positive integer greater than 1. Then if  $n$  is a positive integer, it can be expressed uniquely in the form:

$$n = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b + a_0$$

where  $k$  is a nonnegative integer;

$a_0, a_1, \dots, a_k$  are nonnegative integers less than  $b$ , and  $a_k \neq 0$ .

The  $a_j$  terms with  $j = 0, \dots, k$  are called the **base- $b$  digits** of the representation.

- The representation of  $n$  given in Theorem 1 is called the **base  $b$  expansion of  $n$**  and is denoted by  $(a_k a_{k-1} \dots a_1 a_0)_b$ .

# Base $b$ Representations

From the division algorithm, we can write

$$n = b \cdot q_0 + r_0 ; \text{ where } 0 \leq r_0 < b \text{ and } q_0 < n.$$

Now,  $q_0$  can also be written as:  $q_0 = b \cdot q_1 + r_1$ , where  $q_1 < q_0$ .

and consequently,  $q_1 = b \cdot q_2 + r_2$ , where  $q_2 < q_1$ , and so on.

Eventually, after  $k$  steps,  $q_k$  will be zero . So,  $q_{k-1} = b \cdot 0 + r_k = r_k$

Thus we have,

$$\begin{aligned} n &= b \cdot q_0 + r_0 = b \cdot (b \cdot q_1 + r_1) + r_0 = b^2 \cdot q_1 + b \cdot r_1 + r_0 \\ &= b^2 \cdot (b \cdot q_2 + r_2) + b \cdot r_1 + r_0 = b^3 \cdot q_2 + b^2 \cdot r_2 + b \cdot r_1 + r_0 \\ &= b^{k+1} \cdot q_k + b^k \cdot r_k + b^{k-1} \cdot r_{k-1} + \dots + b^3 \cdot r_3 + b^2 \cdot r_2 + b \cdot r_1 + r_0 \\ &= b^k \cdot r_k + b^{k-1} \cdot r_{k-1} + \dots + b^3 \cdot r_3 + b^2 \cdot r_2 + b \cdot r_1 + r_0 [q_k = 0] \end{aligned}$$

# Binary Expansions

Most computers represent integers and do arithmetic with binary (base 2) expansions of integers. In these expansions, the only digits used are 0 and 1.

**Example :** What is the decimal expansion of the integer that has  $(1\ 0101\ 1111)_2$  as its binary expansion?

**Solution :**  $(1\ 0101\ 1111)_2$   
 $= 1 \cdot 2^8 + 0 \cdot 2^7 + 1 \cdot 2^6 + 0 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 351.$

**Example :** What is the decimal expansion of the integer that has  $(11011)_2$  as its binary expansion?

**Solution :**  $(11011)_2 = 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 27.$

# Octal Expansions

The octal expansion (base 8) uses the digits  $\{0,1,2,3,4,5,6,7\}$ .

**Example :** What is the decimal expansion of the number with octal expansion  $(7016)_8$ ?

**Solution :**  $7 \cdot 8^3 + 0 \cdot 8^2 + 1 \cdot 8^1 + 6 \cdot 8^0 = 3598$

**Example :** What is the decimal expansion of the number with octal expansion  $(111)_8$ ?

**Solution :**  $1 \cdot 8^2 + 1 \cdot 8^1 + 1 \cdot 8^0 = 64 + 8 + 1 = 73$

# Hexadecimal Expansions

The hexadecimal expansion needs 16 digits, but our decimal system provides only 10. So letters are used for the additional symbols. The hexadecimal system uses the digits {0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F}. The letters A through F represent the decimal numbers 10 through 15.

**Example :** What is the decimal expansion of the number with hexadecimal expansion  $(2AE0B)_{16}$ ?

**Solution :**

$$2 \cdot 16^4 + 10 \cdot 16^3 + 14 \cdot 16^2 + 0 \cdot 16^1 + 11 \cdot 16^0 = 175627$$

**Example :** What is the decimal expansion of the number with hexadecimal expansion  $(E5)_{16}$ ?

**Solution :**  $14 \cdot 16^1 + 5 \cdot 16^0 = 224 + 5 = 229$



# Base Conversion

**Example** : Find the octal expansion of  $(12345)_{10}$

**Solution** : Successively dividing by 8 gives:

$$12345 = 8 \cdot 1543 + 1$$

$$1543 = 8 \cdot 192 + 7$$

$$192 = 8 \cdot 24 + 0$$

$$24 = 8 \cdot 3 + 0$$

$$3 = 8 \cdot 0 + 3$$

The remainders are the digits from right to left, yielding  $(30071)_8$ .

# Comparison of Hexadecimal, Octal, and Binary Representations

**TABLE 1** Hexadecimal, Octal, and Binary Representation of the Integers 0 through 15.

|                    |   |   |    |    |     |     |     |     |      |      |      |      |      |      |      |      |
|--------------------|---|---|----|----|-----|-----|-----|-----|------|------|------|------|------|------|------|------|
| <b>Decimal</b>     | 0 | 1 | 2  | 3  | 4   | 5   | 6   | 7   | 8    | 9    | 10   | 11   | 12   | 13   | 14   | 15   |
| <b>Hexadecimal</b> | 0 | 1 | 2  | 3  | 4   | 5   | 6   | 7   | 8    | 9    | A    | B    | C    | D    | E    | F    |
| <b>Octal</b>       | 0 | 1 | 2  | 3  | 4   | 5   | 6   | 7   | 10   | 11   | 12   | 13   | 14   | 15   | 16   | 17   |
| <b>Binary</b>      | 0 | 1 | 10 | 11 | 100 | 101 | 110 | 111 | 1000 | 1001 | 1010 | 1011 | 1100 | 1101 | 1110 | 1111 |

Initial 0s are not shown

- Each octal digit corresponds to a block of 3 binary digits.
  - Each hexadecimal digit corresponds to a block of 4 binary digits.
- So, conversion between binary, octal, and hexadecimal is easy.

# Conversion Between Binary, Octal, and Hexadecimal Expansions

**Example :** Find the octal and hexadecimal expansions of  $(11\ 1110\ 1011\ 1100)_2$ .

**Solution :**

- To convert to octal, we group the digits into blocks of three, adding initial 0s as needed:

$$(011\ 111\ 010\ 111\ 100)_2$$

The blocks from left to right correspond to the digits 3, 7, 2, 7 and 4. Hence, the solution is  $(37274)_8$ .

- To convert to hexadecimal, we group the digits into blocks of four, adding initial 0s as needed:

$$(0011\ 1110\ 1011\ 1100)_2$$

The blocks from left to right correspond to the digits 3, E, B and C. Hence, the solution is  $(3EBC)_{16}$ .

# Binary (Fast) Modular Exponentiation

In cryptography, it is important to be able to find  $b^n \bmod m$  efficiently, where  $b$ ,  $n$ , and  $m$  are large integers.

- We can compute the exponential using the fact:

$$b^{s+t} \bmod m = (b^s \bmod m)(b^t \bmod m) \bmod m$$

- Also, binary representation of  $n$  yields:

$$n = a_{k-1}2^{k-1} + \dots + a_12 + a_0 = (a_{k-1} \dots a_1 a_0)_2$$

- Note that Here  $a_k, a_{k-1}, \dots, a_1, a_0 \in \{0, 1\}$ .

- Then,  $b^n = b^{a_{k-1} \cdot 2^{k-1} + \dots + a_1 \cdot 2 + a_0} = b^{a_{k-1} \cdot 2^{k-1}} \dots b^{a_1 \cdot 2} \cdot b^{a_0}$

- To calculate  $b^n \bmod m$ , successively find:

$$b \bmod m, b^2 \bmod m, b^4 \bmod m, \dots, b^{2^{k-1}} \bmod m$$

Then, multiply together the terms  $b^{2^j} \bmod m$  where  $a_j = 1$ .

**Example :** Compute  $3^{11}$  using this method.

**Solution :** Note that  $11 = (1011)_2$  so that  $3^{11} = 3^8 3^2 3^1 = ((3^2)^2)^2 3^2 3^1 = (9^2)^2 \cdot 9 \cdot 3 = (81)^2 \cdot 9 \cdot 3 = 6561 \cdot 9 \cdot 3 = 117,147$ .

# Fast Modular Exponentiation Example

**Example :** Find  $3^{144} \bmod 100$ .

**Solution :** Here,  $144 = (10010000)_2$  so  $3^{144} = 3^{128} 3^{16}$ . Now,

- $3 \bmod 100 = 3$
- $3^2 \bmod 100 = (3 * 3) \bmod 100 = 9$
- $3^4 \bmod 100 = (9 * 9) \bmod 100 = 81$
- $3^8 \bmod 100 = (81 * 81) \bmod 100 = 6561 \bmod 100 = 61$
- $3^{16} \bmod 100 = (61 * 61) \bmod 100 = 3721 \bmod 100 = 21$
- $3^{32} \bmod 100 = (21 * 21) \bmod 100 = 441 \bmod 100 = 41$
- $3^{64} \bmod 100 = (41 * 41) \bmod 100 = 1681 \bmod 100 = 81$
- $3^{128} \bmod 100 = (81 * 81) \bmod 100 = 6561 \bmod 100 = 61$

So,  $3^{144} \bmod 100 = (3^{128} \bmod 100)(3^{16} \bmod 100) \bmod 100$   
 $= (61 * 21) \bmod 100 = 1281 \bmod 100 = 81$ .

Therefore,  $3^{144} \bmod 10 = 81$  (Answer)